

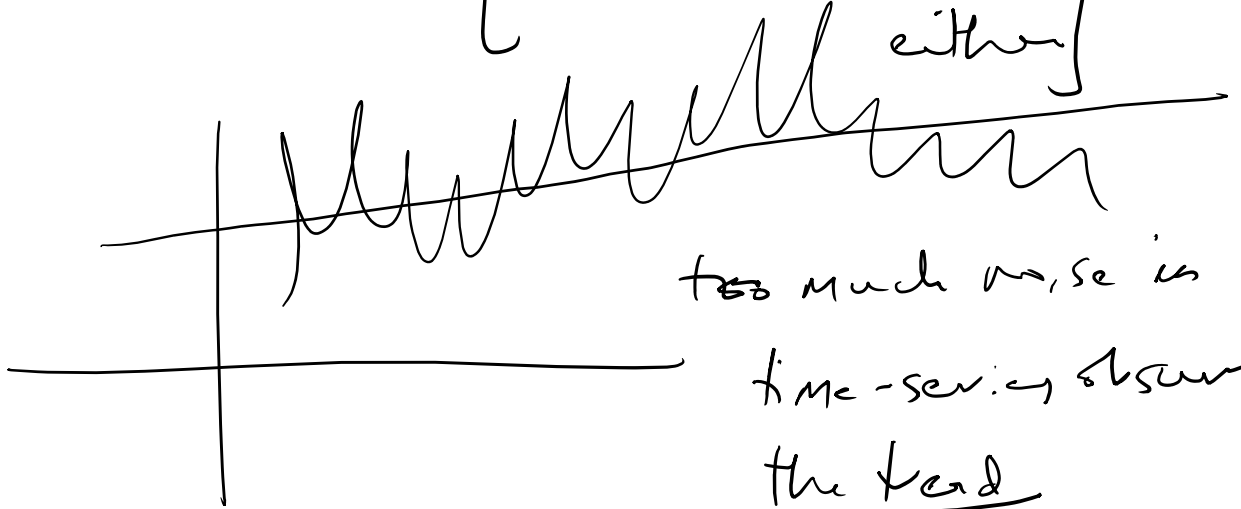
Seasonality $\rightarrow$ short term, regular patterns
Sales increase in December

If we only know Y_t as a function of t ,
how do we know which model to
apply? [We do not know for sure]
Additive or Multiplicative

Why not this?

$$Y_t = T_t + S_t + C_t + I \quad (?)$$

[we do not know this either]



too much noise is

time-series obscures

the trend

y_t has units of kgs

$$\textcircled{y_t} = \textcircled{T_t} \times \textcircled{S_t} \times \textcircled{C_t} \times \textcircled{I_t} \quad [\text{Multiplicative model}]$$

$\uparrow$ kgs $\downarrow$ kgs $\downarrow$ 1.2 $\downarrow$ 0.8 $\downarrow$ 1.5

In an additive model $\Delta y_t = S_t$

$$\Delta y_t = y_t - T_t$$

$$y_t = \textcircled{T_t} + \textcircled{S_t}$$

$$y_t = T_t S_t \quad (\text{basic multiplicative model})$$

$$y_t - T_t = T_t S_t - T_t = T_t (S_t - 1)$$

$$\frac{y_t - T_t}{T_t} = \frac{T_t S_t - T_t}{T_t} = (S_t - 1)$$

Relative deviation is a constant in
a multiplicative model.

Absolute deviation is a constant in
an additive model

What is Delphi method for forecasting?

expert (E1) (E2) ... (E10)

circulate a questionnaire
predict how many EVs will be sold in
5 years time.

E1 $\rightarrow$ expert in battery tech

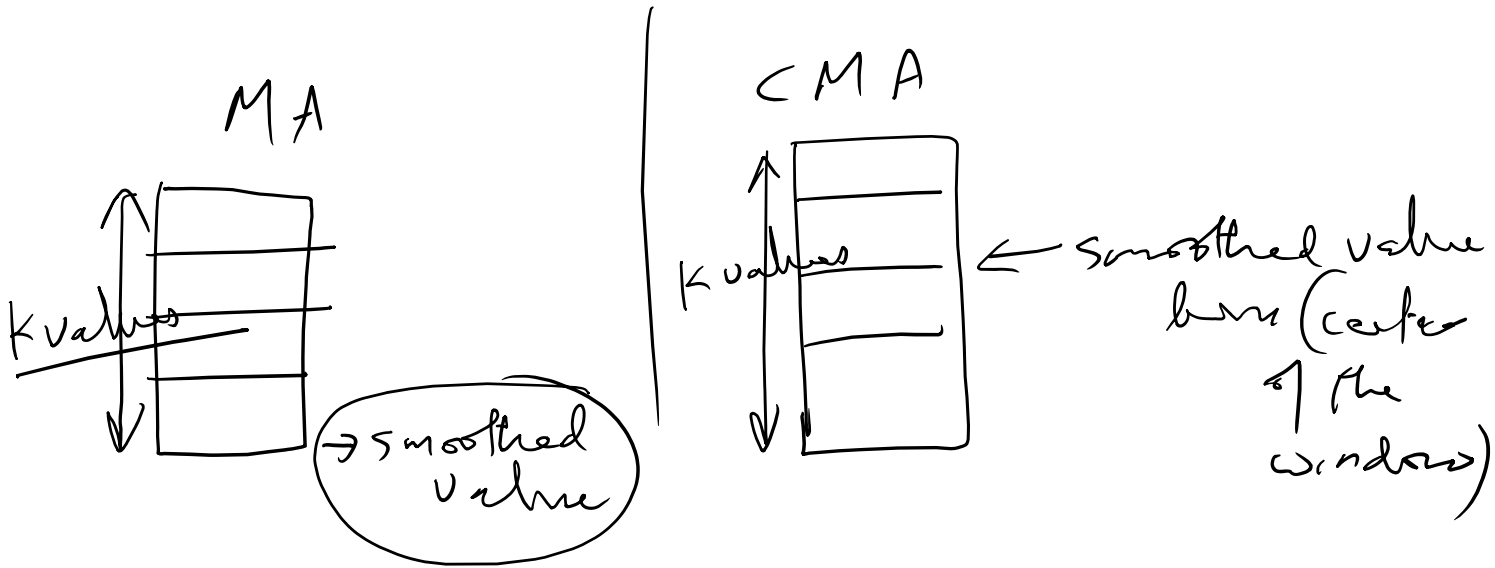
E2 $\rightarrow$ expert in government policy

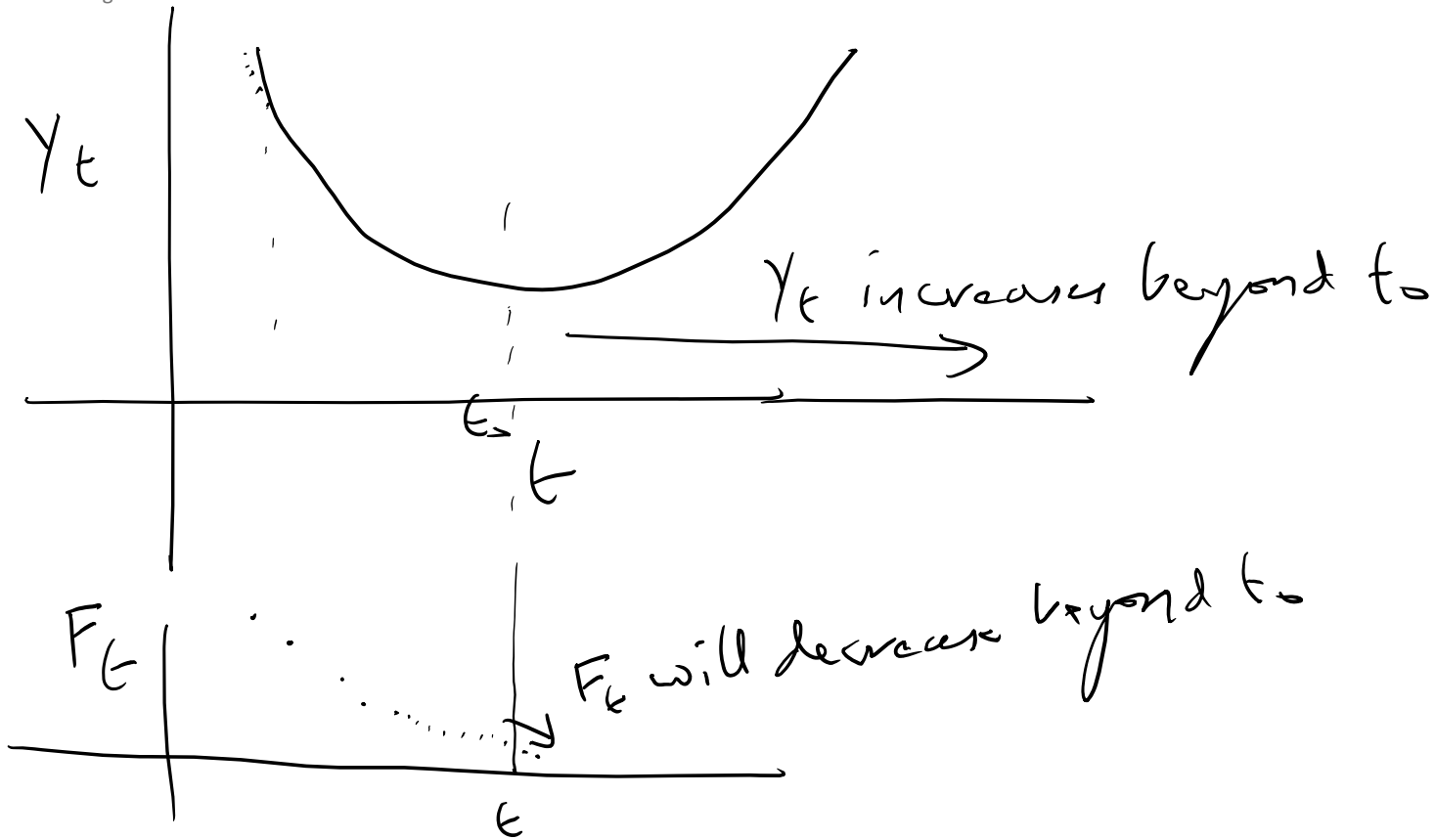
What is the difference between time-series
& causality?

$y_1, y_2, \dots, y_t$ } time-series
(sales is a function
only of previous sales)

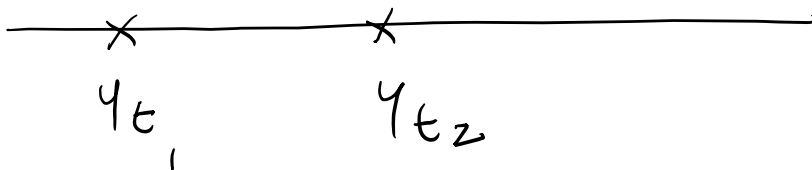
Causality $\rightarrow$ build a model $\rightarrow$ do version

$$\text{Sales} = f(\text{advertising expenditure}, \text{product quality})$$





$$f(y_{t_1}, y_{t_2}) = f(y_{t_1+h}, y_{t_2+h})$$



y_{t_1} and y_{t_2} are both random variables

$f(y_{t_1}, y_{t_2}) \rightarrow$ asking how do these random variables behave together?

$$P(y_{t_1} = 10, y_{t_2} = 20) = \underline{0.01}$$

Suppose $t_1 = t_1 + \Delta$ and $t_2 = t_2 + \Delta$, so
 we have two new points and two new
 random variables $y_{t_1 + \Delta}, y_{t_2 + \Delta}$

$$P(y_{t_1 + \Delta} = 10, y_{t_2 + \Delta} = 20) = 0.01$$

Joint probability distribution table for y_{t_1}, y_{t_2} is the same as joint probability distribution table for $y_{t_1+\Delta}, y_{t_2+\Delta}$

$$\{Y_t\}$$

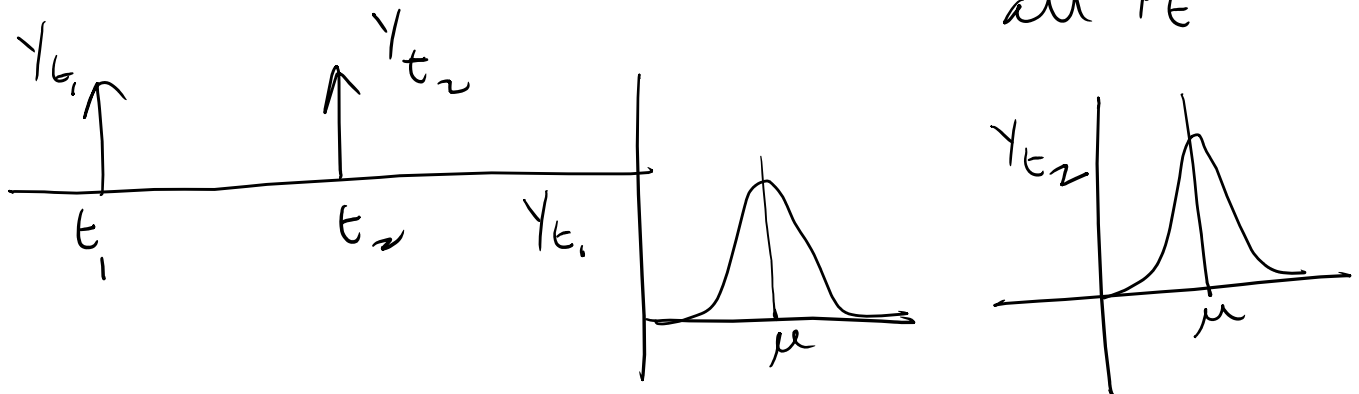
$$E[Y_t] = \mu_{Y_t} = \mu \text{ (because of stationarity)}$$

$$f(y_{t_1}, y_{t_1+h}) = f(y_{t_2}, y_{t_2+h})$$

$$\text{Take } h = 0$$

$$f(y_{t_1}, y_{t_1}) = f(y_{t_2}, y_{t_2})$$

$$\boxed{f(y_{t_1})} = f(y_{t_2}) \quad \left. \begin{array}{l} \text{distribution of } Y_t \\ \text{is the same for} \\ \text{all } Y_t \end{array} \right\}$$



Distribution of Y_{t_1} and Y_{t_2} is the same
 $\Rightarrow$ Mean of $Y_{t_1} = \text{mean of } Y_{t_2} = \mu$

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\text{Cov}(Y_t, Y_{t+h}) = E[(Y_t - \mu)(Y_{t+h} - \mu)]$$

Since $\mu_{Y_t} = \mu_{Y_{t+h}} = \mu$

$$E[(Y_t - \mu)(Y_{t+h} - \mu)]$$

$$= E[Y_t Y_{t+h} - \underbrace{Y_t \mu}_{\downarrow} - \underbrace{\mu Y_{t+h}}_{\downarrow} + \mu^2]$$

$$= E[Y_t Y_{t+h}] - \underbrace{\mu E[Y_t]}_{\downarrow} - \mu E[Y_{t+h}] + \mu^2$$

$$= E[Y_t Y_{t+h}] - \mu^2 - \mu^2 + \mu^2$$

$$\text{Cov}(Y_t, Y_{t+h}) = E[Y_t Y_{t+h}] - \mu^2$$

$$E[Y_t Y_{t+h}] = \iint Y_t Y_{t+h} \underbrace{f(Y_t, Y_{t+h})}_{\text{joint density}} dY_t dY_{t+h}$$

this integral is the same for all values of t since $f(Y_t, Y_{t+h})$ is the same for all values of t

$\therefore E[Y_t Y_{t+h}]$ depends only on h

$$\therefore \boxed{\text{Cov}(Y_t, Y_{t+h}) = \gamma_h}$$

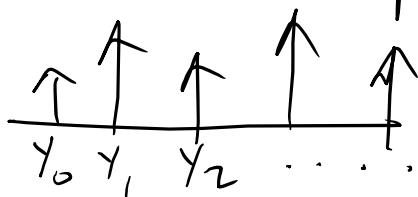
$$E[Y_1 Y_2] = E[Y_3 Y_4] = E[Y_{100} Y_{101}] = \gamma_1$$

$$r_k = \frac{\text{Cov}(Y_t, Y_{t+k})}{\text{Cov}(Y_t, Y_t)}$$

Time series



Multipl. realizations of the time series needed
to compute r_k



formula in the slides

$$= \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$

is only one-realization of the time series

The two quantities (v_k across multiple realizations) and v_k across a single realization are the same because of the property of ergodicity

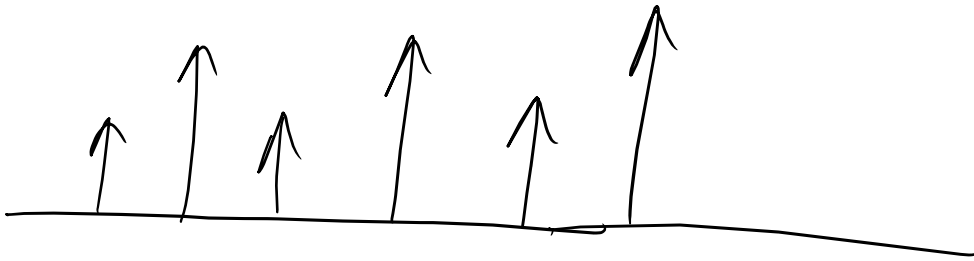
White noise time series

$$X_t \sim WN(0, \sigma^2)$$

$$\bar{E}[X_t] = 0$$

$$Var[X_t] = \sigma^2$$

$$Cov(X_t, X_{t-k}) = 0 \text{ for every } k \neq 0$$



totally independent spikes at each time instant

$$\hat{\rho}_k = \frac{\sum_{t=1}^{T-k} (X_t - \bar{X})(X_{t+k} - \bar{X})}{\sum_{t=1}^T (X_t - \bar{X})^2}$$

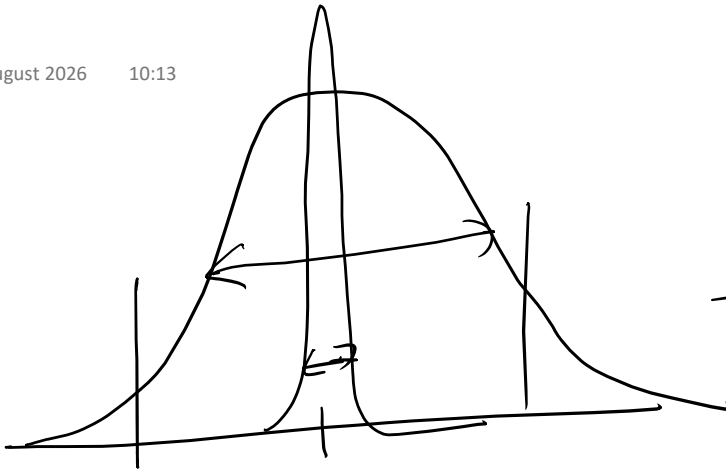
numerator is an average of many products of independent random variables
 By CLT it is approximately normal

The denominator is the sample variance and if T is large enough, the denominator is $T\sigma^2$

After dividing by sample variance

$$\hat{\beta}_k \approx N\left(\underset{\text{mean } 0}{0}, \underset{\text{variance } \frac{1}{T}}{\frac{1}{T}}\right)$$

for fixed k when T is reasonably large



$$\hat{\rho}_k \text{ dist}$$

Since $\hat{\rho}_k \approx N\left(0, \frac{1}{T}\right)$ 95% of the observations will lie within ± 1.96 std deviations of the mean

$\hat{\rho}_k$ will lie with 95% probability in the range $\left(-\frac{1.96}{\sqrt{T}}, +\frac{1.96}{\sqrt{T}}\right)$